

Alex Spigler

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EDUCATION

George Washington University

Dual Degree: BS in Statistics, BA in Computer Science; Minor in Mathematics

GPA: 3.77 | Statistics GPA: 3.88

Expected May 2027

Washington, DC

EXPERIENCE

Data Analyst Intern

AnswersNow

Jul. – Aug. 2026

Remote

- Authored a data-rules specification defining ~25 client and utilization metrics at the warehouse column and flag level, resolving a cross-team cancellation-rate discrepancy caused by tentative schedule records and standardizing executive-reporting definitions.
- Proposed and completed a manager-approved competing-risks analysis of 2,121 clients using cause-specific Cox models; a 10-percentage-point higher Stage-1 attendance rate was associated with lower dropout hazard at day 90 (HR 0.840) and day 365 (HR 0.936), with time-varying and influence diagnostics.
- Analyzed 23,406 expected client-weeks across 1,232 clients in a production PostgreSQL warehouse, integrating billing, appointment, authorization, and payer data into a 39-chart Mode report that found the weekly share receiving therapy fell from 86–89% to 78–81% while monthly reach held near 88%.

Barista

Tatte Bakery & Cafe

Feb. 2023 – Nov. 2024

Washington, DC

- Trained 7 new employees on store operations and customer service after consistently reliable performance.

PROJECTS

U.S. Unemployment Rate Forecasting | *R*

- Built a reproducible forecasting study on 942 observed monthly BLS unemployment rates (1948–2026), preserving a source-documented missing month and reserving the final 24 calendar months as holdout.
- Compared 24 diagnostically specified SARIMA models and ETS across 10 rolling windows; on a held-out 24-month period, the selected model beat naive and drift on MAE but not RMSE, exposing regime sensitivity.

Chaotic Attractors: Computational Exploration & Visualization | *Python*

- Built a pip-installable Python library (CLI + API) for searching, validating, and visualizing 9 four-parameter 2D systems (5 established forms, 4 custom variants).
- Designed a seeded, staged search that removes burn-in and validates survivors with finite-time largest Lyapunov estimates; ranked outputs by geometry and rendered density with subsampled Gaussian KDE.

TECHNICAL SKILLS

Languages:	Python, R, SQL, Java
Statistical Methods:	Experimental Design & Power Analysis, Bayesian Inference & MCMC, Regression Diagnostics, Survey Sampling & Weighting, Time Series & Intervention Analysis, Survival Analysis & Competing Risks
Mathematical Methods:	Linear Algebra, Dynamical Systems & Stability Analysis, Numerical ODE Methods
Libraries & Tools:	NumPy, pandas, SciPy, Matplotlib, Git, pytest, Mode Analytics, R Markdown
Databases:	PostgreSQL